Question 1: Basic PHP Syntax and Functions

1.1 (2 points): Write a PHP script that declares three variables:

- studentName (your full name as a string)
- studentAge (your age as an integer)
- studyYear (your current year of study as an integer)

Then, use these variables to output a sentence in the following format: "My name is [studentName], I am [studentAge] years old, and I am in year [studyYear] of my studies."

1.2 (2 points): Create a PHP function named calculateCircleArea that takes one parameter: radius (float). The function should return the area of a circle using the formula: **Area = π × radius²** (use M_PI constant).

Demonstrate the function by calculating and displaying the area when radius = 7.5. Output format example: "The area of the circle with radius 7.5 is: 176.71458676443"

Question 2: Arrays and Control Structures

2.1 Create an indexed array named $cities containing at least 5 Vietnamese city names. Use a **foreach** loop to display each city name inside an HTML unordered list (<ul><li>...</li></ul>).

Example output:

```
<ul>
    <li>Hanoi</li>
    <li>Ho Chi Minh City</li>
    ...
</ul>
```

Given the following associative array:

```
$scores = [
    'Math'     => 8.5,
    'Physics'  => 7.0,
    'English'  => 9.2,
    'History'  => 6.8
];
```

Write a PHP script that:

- Increases every score by 0.5 (but not exceeding 10.0)
- Outputs a table showing subject and new score.

Example output:

```
Subject     | New Score
-----------|----------
Math        | 9.0
Physics     | 7.5
English     | 9.7
History     | 7.3
```

2.2 Array of Students

Create an **indexed array** named $students containing 4 associative arrays. Each associative array represents a student with the following keys:

- id
- name
- age
- grade

Example data:

PHP

```php
$students = [
    ['id' => 'SV001', 'name' => 'Nguyen Van A', 'age' => 20, 'grade' => 8.5],
    // Thêm 3 sinh viên khác
];
```

Using a foreach loop, display all students in an HTML table with columns: ID, Name, Age, Grade.

Question 3: Class & Object

3.1 Create Student Class (3 Points)

Create a class named Student with the following:

- Private properties: $id, $name, $age, $grade

- Constructor that accepts all 4 parameters
- Getters for all properties
- A method getRank() that returns:
  - "Excellent" if grade ≥ 9.0
  - "Good" if grade ≥ 7.0
  - "Average" if grade ≥ 5.0
  - "Weak" otherwise

3.3 Array of Objects (2 Points)

Convert the array from 3.1 into an **array of Student objects**. Display the list again using the getRank() method in an extra column.

UI Mockup:

text

```
-------------------------------------------------------
| Student List                                         |
-------------------------------------------------------
| ID    | Name           | Age | Grade | Rank        |
| SV001 | Nguyen Van A   | 20  | 8.5   | Good        |
| SV002 | Tran Thi B     | 19  | 9.2   | Excellent   |
-------------------------------------------------------
```

Question 4: Product Object with Input/Output

Objective: Create and manage **Product objects** with user input and display.

4.1 Create Product Class (3 Points)

Create a class named Product with:

- Private properties: $code, $name, $price, $stock
- Constructor
- Getters and Setters for all properties
- Method isInStock() that returns true if $stock > 0
- Method getInfo() that returns a formatted string: "Code: [code] - Name: [name] - Price: [price] VND - Stock: [stock]"

4.2 Form to Add Product (3 Points)

Create an HTML form that allows the user to input:

- Product Code
- Product Name
- Price
- Stock Quantity

When the form is submitted:

- Create a new Product object
- Store the object in a session array $products
- Display a success message

4.3 Display All Products (2 Points)

Below the form, display all products from the session array in a table. Show "In Stock" or "Out of Stock" based on the isInStock() method.

UI Mockup:

text

```
-------------------------------------------------------
| Add New Product                                     |
-------------------------------------------------------
| Code: [________]                                    |
| Name: [________________________]                    |
| Price: [________] VND                               |
| Stock: [________]                                   |
| [ Add Product ]                                     |
-------------------------------------------------------


Product List:
-------------------------------------------------------
| Code | Name          | Price     | Stock | Status    |
| P001 | iPhone 14     | 25,000,000| 12    | In Stock  |
-------------------------------------------------------
```

Question 5: Form Handling, Data Validation & File Upload

4.1 (2 points): Create an HTML form that allows a user to input the following information:

- Full Name (text)
- Email (email)
- Profile Picture (file upload – only jpg, jpeg, png allowed, max 2MB)
- Submit button ("Upload Profile")

The form should submit to the same PHP page using **POST** method and **enctype="multipart/form-data"**.

4.2 (2 points): On form submission, perform the following validations and actions:

- Full Name must not be empty and must contain at least 5 characters.
- Email must be a valid email format.
- Profile picture must be uploaded, must be jpg/jpeg/png, and size ≤ 2MB.
- If all validations pass, save the uploaded file to a folder named uploads/ with a unique name (e.g., timestamp + original name).
- Display a success message including the uploaded file name.
- If any validation fails, display appropriate error message(s) in red.